DSA Mock 1

Total points 38/40

12.Do both of these statements create a string? * 1/1

```
String str = new String("abc");

//or

String str1 = "abc";
```

- (•) Yes ✓
- ( ) NO
- ( ) First is correct
- ( ) Second is correct

. ... form of access is used to add and remove nodes from a queue. * 1/1

- ( ) LIFO, Last In First Out
- (•) FIFO, First In First Out ✓
- ( ) Both a and b
- ( ) None of these

PN *

1.Which of the following is a linear data structure? * 1/1

- (x) Array ✓
- ( ) AVL Trees
- ( ) Binary Trees
- ( ) Graphs

18. What is the maximum number of edges possible in a directed graph having 8 vertices and no self-loops? * 1/1

- ( ) 60
- ( ) 30
- (x) 56 ✓
- ( ) 216

22. Which of the following is the time complexity if the graph is represented as an adjacency matrix in kruskal's algorithm? * 1/1

- (x) O(E log V) ✓
- ( ) O(V log E)
- ( ) O(V^2)
- ( ) O(log E)

39. Consider an implementation of unsorted singly linked list. Suppose it has its representation with a head and tail pointer. Given the representation, which of the following operation can be implemented in O(1) time? * 0/1

i) Insertion at the front of the linked list
ii) Insertion at the end of the linked list
iii) Deletion of the front node of the linked list
iv) Deletion of the last node of the linked list

- I and II
- I and III
- I ,II and III
- I,II and IV

Correct answer

- I and III

26.The following formular is of *
left_subtree (keys) ≤ node (key) ≤ right_subtree (keys) 1/1

- Bianry Tree
- Complete Binary Tree
- Binary Search tree
- All of the above

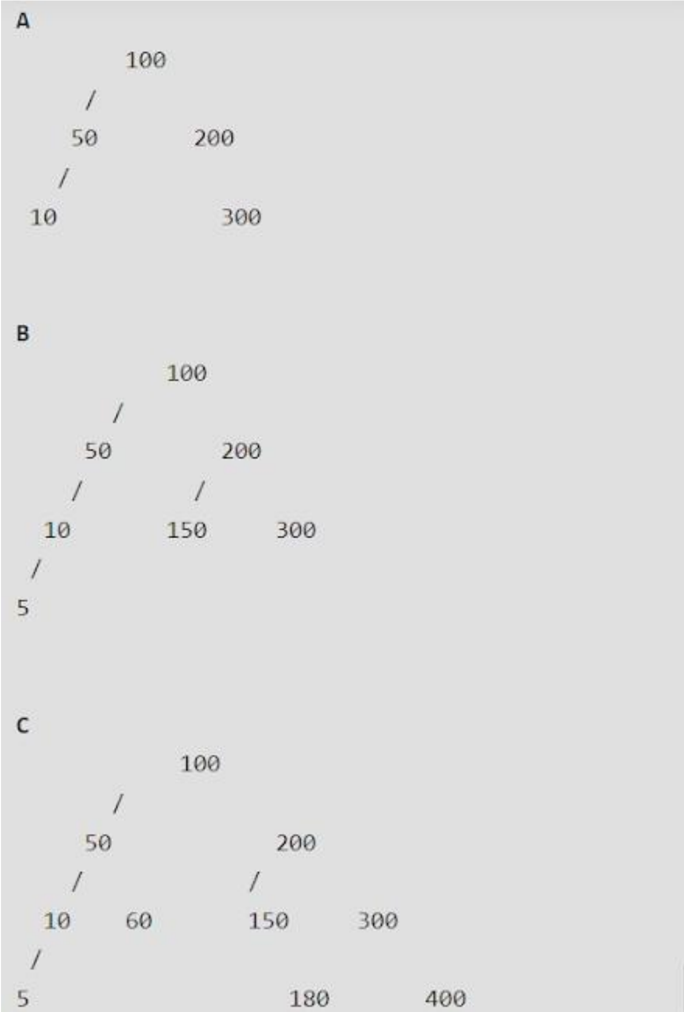
13.Find the Big O complexity of the following code: * 1/1

```
class NestedLoop {
  public static void main(String[] args) {
    int n = 10; // 1 step --> Note: n can be anything. This is just an example
    int sum = 0; // 1 step
    double pie = 3.14; // 1 step

    for (int var = 0; var < n; var = var + 3) { // n/3 steps
      System.out.println("Pie: " + pie); // n/3 steps
      for (int j = 0; j < n; j = j + 2) {  // (n/3 * n/2) steps
        sum++;   // (n/3 * n/2) steps
        System.out.println("Sum = " + sum);  // (n/3 * n/2) steps
      }
    }
  }
}
```

- (•) O(n^2) ✓
- ( ) O(n)
- ( ) n(O)
- ( ) O^2(n)

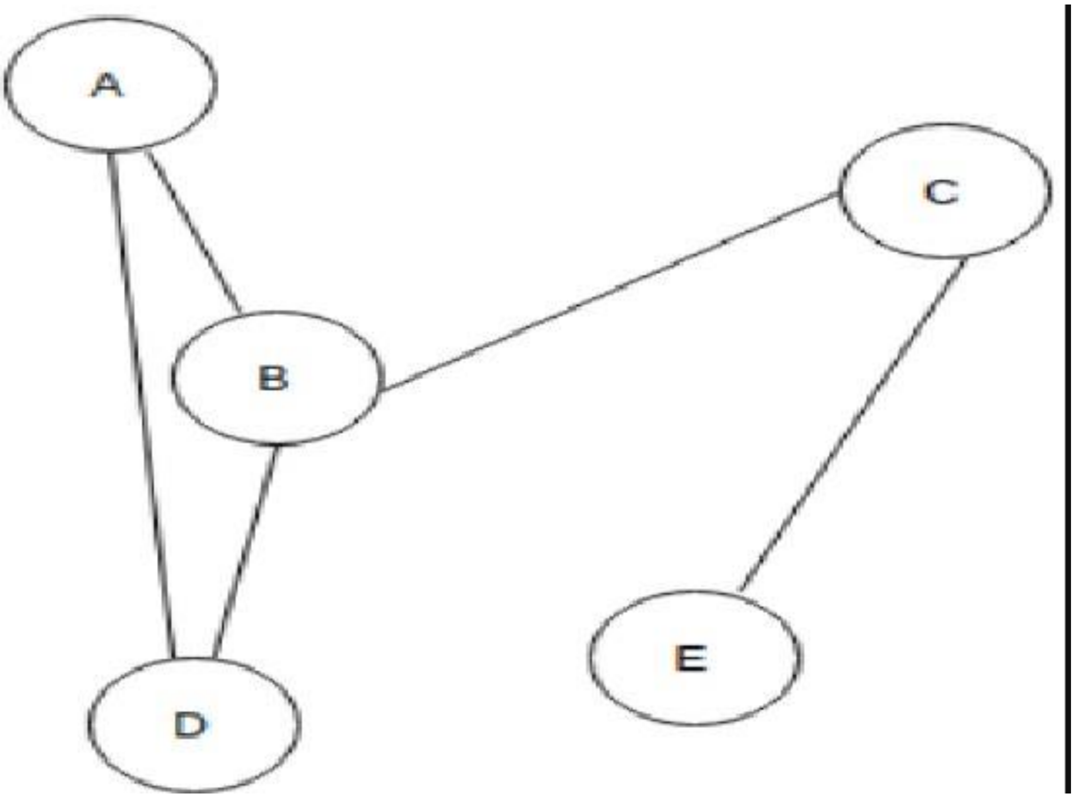
6.Which of the following is AVL Tree * 1/1



- A
- A and C
- A,B and C
- B only

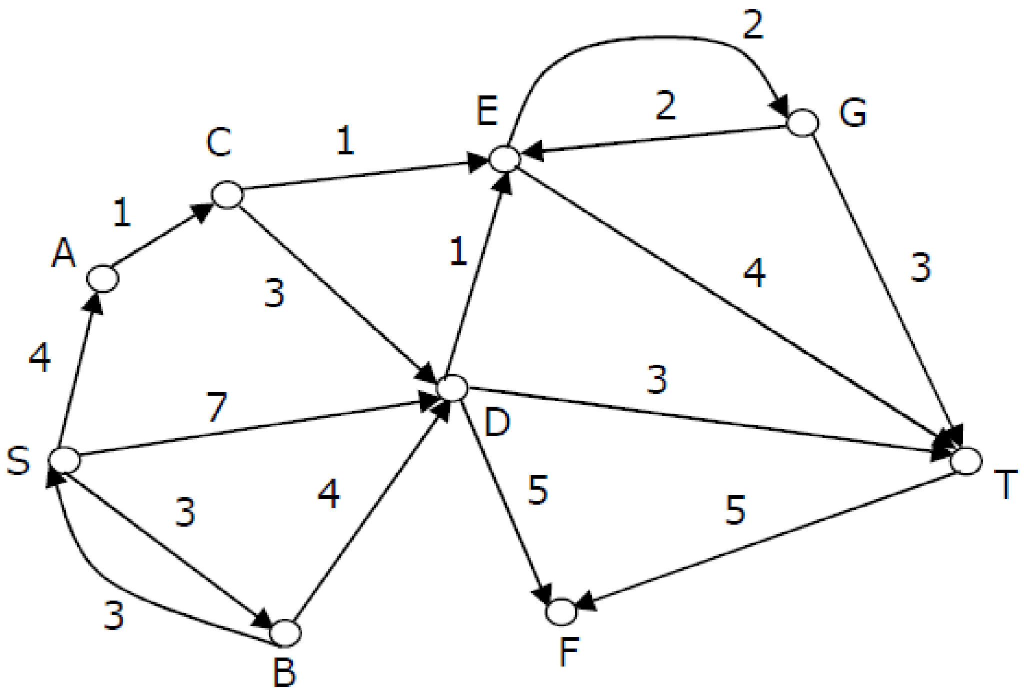
14. In the given graph identify the cut vertices *

1/1



- B and E
- C and D
- A and E
- C and B

8. Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijstra?s shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered. *1/1



- SDT
- SBDT
- SACDT
- SACET (selected)

38.Which of the following option is not correct? * 1/1

- A-If the queue is implemented with a linked list, keeping track of a front pointer, Only rear pointer s will change during an insertion into an non-empty queue.
- B-Queue data structure can be used to implement least recently used (LRU) page fault algorithm and Quick short algorithm.
- C-Queue data structure can be used to implement Quick short algorithm but not least recently used (LRU) page fault algorithm.
- Both (A) and (C)

37.Consider the following statements: * 1/1

i. First-in-first out types of computations are efficiently supported by STACKS.

ii. Implementing LISTS on linked lists is more efficient than implementing LISTS on an array for almost all the basic LIST operations.

iii. Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices.

iv. Last-in-first-out type of computations are efficiently supported by QUEUES.

- (ii) is true
- (i) and (ii) are true
- (iii) is true
- (ii) and (iv) are true

17. What is the minimum number of queues required for implementing a stack? * 1/1

- 1
- 2 (selected, correct)
- 3
- 4

25.State True or False. * 1/1
i) Binary search is used for searching in a sorted array.
ii) The time complexity of binary search is O(log n).

- True, False
- False ,True
- False ,False
- True ,True (selected, correct)

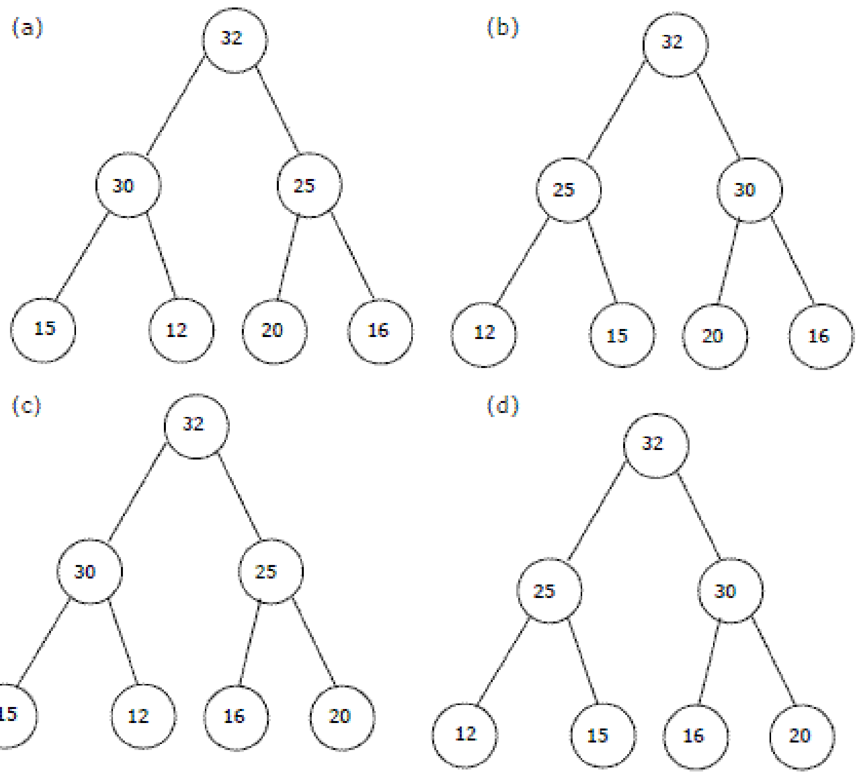
9. A connected planar graph having 6 vertices, 7 edges contains ____________regions * 1/1

- 15
- 7
- 3 (selected, correct)
- 1

33. In a leftist heap, all the operations should be performed on? * 1/1

- left path
- centre path
- root
- right path

The elements 32, 15, 20, 30, 12, 25, 16 are inserted one by one in the given order into a Max Heap. The resultant Max Heap is. *1/1



- a
- b
- c
- d

36. Consider the following operation along with Enqueue and Dequeue operations on queues, where k is a global parameter.
What is the worst case time complexity of a sequence of n MultiDequeue() operations on an initially empty queue?
(A) \Theta(n) (B) \Theta(n + k) (C) \Theta(nk) (D) \Theta(n^2) *1/1

```
MultiDequeue(Q){
   m = k
   while (Q is not empty and m  > 0) {
      Dequeue(Q)
      m = m - 1
   }
}
```

- (●) A ✓
- ( ) B
- ( ) C
- ( ) D

15.Which one of the following is an application of Queue Data Structure? * 0/1

- ( ) When a resource is shared among multiple consumers
- ( ) When data is transferred asynchronously (data not necessarily received at same rate as sent) between two processes
- ( ) Load Balancing
- (●) All of the above ✗

Correct answer

- (●) When data is transferred asynchronously (data not necessarily received at same rate as sent) between two processes

3.What will be the output of the following code snippet 1/1

```
import java.util.*;

public class Main {
   public static void main(String[] args) {
      solve();
   }

   public static void solve() {
      int n = 24;
      int l = 0, r = 100, ans = n;
      while (l <= r) {
         int mid = (l + r) / 2;
         if (mid * mid <= n) {
            ans = mid;
            l = mid + 1;
         } else {
            r = mid - 1;
         }
      }
      System.out.println(ans);
   }
}
```

- 5
- 4 (selected, correct)
- 6
- 3

35.{ * 1/1

```
{
  Stack S; // Say it creates an empty stack S

  // Run while Q is not empty
  while (!isEmpty(Q))
  {
    // deQueue an item from Q and push the dequeued item to S
    push(&S, deQueue(Q));
  }

  // Run while Stack S is not empty
  while (!isEmpty(&S))
  {
   // Pop an item from S and enqueue the popped item to Q
   enQueue(Q, pop(&S));
  }
}
```

- ○ Removes the last from Q
- ○ Keeps the Q same as it was before the call
- ○ Makes Q empty
- ◉ Reverses the Q ✓

2.Which of the following is not the type of queue? * 1/1

- ○ Priority queue
- ◉ Single-ended queue ✓
- ○ Circular queue
- ○ Ordinary queue

11What does the following fragment of code do? * 1/1

```
Node currentNode = list.headNode;
while(currentNode != null){
    currentNode = currentNode.nextNode;
}
```

- (selected) Traverse ✓
- sort
- Hash
- none of the above

4.Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in-order traversal sequence of the resultant tree? 1/1

- 7510324689
- 0243165987
- (selected) 0123456789 ✓
- 9864230157

20.In a circular queue the value of r will be * 1/1

- r=r+1
- (selected) r=(r+1)%QUEUE_SIZE ✓
- r=(r+1)%QUEUE_SIZE-1
- r=(r-1)%QUEUE_SIZE

29. A graph is a tree if and only if graph is * 1/1

- Directed graph
- Contains no cycles
- Planar
- Completely connected

40. In worst case, the number of comparison need to search a singly linked list of length n for a given element is *1/1

- log n
- n/2
- log2n-1
- n

31. Which of the following is the infix expression? * 1/1

- ABC+*
- A+B*C
- +A*BC
- None of the above

23.Below program that takes a number as an argument, and uses a stack S to do processing. *1/1

```
public void convert( int no ) {
Stack St; // For creating the empty stack
while ( no > 0 ) {
// For pushing the value of no%2 into stack St
push( St, no%2 );
no = no/2;
}
// Continuously checking whether stack is empty or not
while ( !isEmpty( St ) ) {
System.out.println( "%d ", pop( St )); // pop an element from St and print it
} }
```

- ○ Prints binary representation of number in reverse order
- ◉ Prints binary representation of number ✓
- ○ Prints the value of number
- ○ Prints the value of number in reverse order

10. What would you enter in the blank to run through all elements of the array? *1/1

```
int arr[] = {1,2,3,4};

for (int i = 0; i < ______ ; i++){

    System.out.println(arr[i]);

}
```

- (•) arr.length ✓
- ( ) arr.length()
- ( ) arr.size()
- ( ) arr.size

7. **Which of the following sorting algorithms is preferred to sort a linked list?** *1/1

- (•) Merge Sort ✓
- ( ) Quick Sort
- ( ) Insertion Sort
- ( ) None of the Above

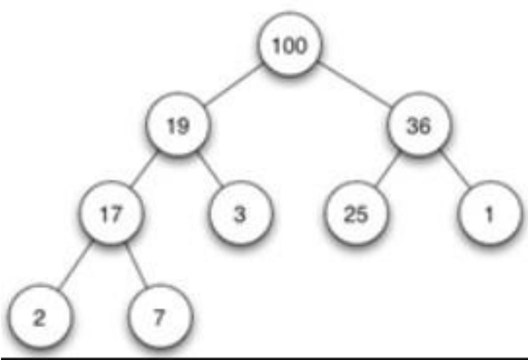
5.Suppose a circular queue of capacity (n – 1) elements is implemented with an array of n elements. Assume that the insertion and deletion operation are carried out using REAR and FRONT as array index variables, respectively. Initially, REAR = FRONT = 0. The conditions to detect queue full and queue empty are

1/1

- (selected) Full: (REAR+1) mod n == FRONT, empty: REAR == FRONT ✓
- Full: (REAR+1) mod n == FRONT, empty: (FRONT+1) mod n == REAR
- Full: REAR == FRONT, empty: (REAR+1) mod n == FRONT
- Full: (FRONT+1) mod n == REAR, empty: REAR == FRONT

32. . If we implement heap as maximum heap , adding a new node of value 15 to the left most node of right subtree. What value will be at leaf nodes of the right subtree of the heap.

*1/1



- (selected) 15 and 1 ✓
- 25 and 1
- 3 and 1
- 2 and 3

21. The result of prefix expression * / b + - d a c d, where a = 3, b = 6, c = 1, d = 5 is * 1/1

- ( ) 0
- ( ) 5
- (•) 10 ✓
- ( ) 8

Name *

....

34. choose the output * 1/1

```
package demo;

public class DemoTranslation {
	public static void main(String[] args) {
		int i = lAbc(10);
		System.out.println(--i);
	}

	public static int lAbc(int i) {
		return i++;
	}
}
```

- ( ) 10
- ( ) 11
- (•) 9 ✓
- ( ) 8

16.A hash table of length 10 uses open addressing with hash function h(k)=k mod 10, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below. *1/1

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

- ( ) 46, 42, 34, 52, 23, 33
- ( ) 34, 42, 23, 52, 33, 46
- (•) 46, 34, 42, 23, 52, 33 ✓
- ( ) 42, 46, 33, 23, 34, 52

19. What is the condition of the overflow of a linear queue implemented using an array? *1/1

- ( ) Rear = front + 1
- ( ) Rear = front
- ( ) Rear = MAX_SIZE
- (•) Rear = MAX_SIZE - 1 ✓

24.What will be the running-time of Dijkstra's single source shortest path algorithm, if the graph G(V,E) is stored in form of adjacency list and binary heap is used 1/1

- O(IVI log IVI)
- O(IVI^2)
- O(IEI +IVI log IVI)
- None of these

28. A full binary tree with n leaves contains * 1/1

- n - 1 nodes
- log2n nodes
- 2n – 1 nodes
- 2^n nodes